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Completion and Perforation Strategy in Unconventional Thin Laminated Interbedded Shales- Case History in Pakistan

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Abstract

The success of fracturing in an unconventional reservoir is largely dependent upon the reservoir characteristics and the frac design. Hence, a holistic analysis of the field encompassing both subsurface and operational parameters and thereby evaluating potential risks, is critical in determining a candidate well for a successful hydraulic frac job. In the unconventional reservoir, Frac job is a critical factor for the success of the project. In Pakistan the unconventional is a new arena. In this paper, two frac designs are compared, a conventional approach and a tailored fit-for-the-purpose frac design. The difference between the production gains is tremendous despite having comparable volumes of proppant and fluids pumped in the two cases.

Geomechanical and petrophysical assessments were done followed by reservoir and production engineering analysis. A workflow was developed that includes building of 1D MEM followed by a fracture design, well performance modelling to estimate incremental production including sensitivity analysis for fracture geometry and number of stages, economic evaluation to identify the most effective completion design. The workflow has overcome the difficulty for comparing frac designs and pumping schedules in terms of production rates and pressures in thin laminated beds. This workflow integrates injection and production data to correlate and select the optimum design for execution and estimation of production gains

This workflow was applied to new wells to identify the optimum design. The well performance is aligned with the results of the model. In thin, laminated reservoirs, achieving effective stimulation requires more than just the successful execution of a hydraulic fracturing operation.

This case study illustrates that a carefully engineered fracture schedule designed to minimize height recession and promote either dense proppant packing or a controlled screen out can lead to significant improvements in well performance, even in the geologically complex thin-bed formations.

This was done on first interbedded shale play in Pakistan and would serve as a guideline for the future jobs. This resulted in 3-4 folds increase in production.

Introduction

The Ghazij Gas discovery is in the Daharki District of Sindh Province, Pakistan, producing gas from thin limestone intervals within the Eocene-aged Ghazij Formation. This formation has an average gross thickness of approximately 230 meters (755 ft) and consists of interbedded clay-rich shales and low-clay

limestones, with individual limestone beds ranging from 1 to 3 meters (3-10 ft) in thickness. Despite their thin depositional nature, these gas-bearing limestones exhibit lateral continuity and can be correlated across the 30-kilometer diameter of the field. The discovery was confirmed with the drilling of the Ghazij-1 well in 2022, reaching a depth of 1015 mTVD. Subsequent appraisal drilling included nine additional wells, all of which tested positive for gas within the Ghazij main fault block.

Petrophysical Challenges and Advanced Evaluation Techniques

The Ghazij formation presents several petrophysical evaluation challenges due to its thin-bedded reservoir units and resistivity suppression in its limestone intervals. Conventional wireline logging tools struggle to accurately measure properties in such thin beds, leading to uncertainties in shale volume and water saturation calculations. To address these issues, advanced logging suites, including Nuclear Magnetic Resonance (NMR) logs, have been deployed. NMR data provides high-resolution, lithology-independent porosity measurements, mitigating uncertainties associated with neutron-density logs. However, resistivity measurements remain problematic due to shoulder bed effects from adjacent shale layers, which suppress resistivity readings such as an average log-derived water saturation of 88%, despite wells producing dry gas. To improve the accuracy, advanced logging techniques were used in some wells which include LWD-Resistivity in deviated wells and RT-Scanner in vertical wells to improve reservoir characterization. Formation testing via XPT (Extended Pressure Testing) has yielded limited success, with only 20 out of 115 pretests providing stable results. However, Production logging (PSP) has been very useful in assessing individual lobe contributions, revealing significant variability in flow rates.

Reservoir Characterization and Lobe-Based Correlation

The operator has adopted a lobe-based classification system to subdivide the Ghazij formation into laterally continuous limestone intervals. The Ghazij reservoir consists of over 31 thin limestone stringers isolated by shale or claystone layers. While some lobes consist of single limestone beds, others comprise of multiple closely-spaced beds with similar depositional and reservoir characteristics. Production performance varies significantly among the lobes, with some having substantial gas flow contribution while others exhibit minimal or no contribution. Permeability heterogeneity is the key factor influencing productivity, with certain lobes behaving as tight gas zones. Hydraulic fracturing is being considered to enhance production from these low-permeability intervals.

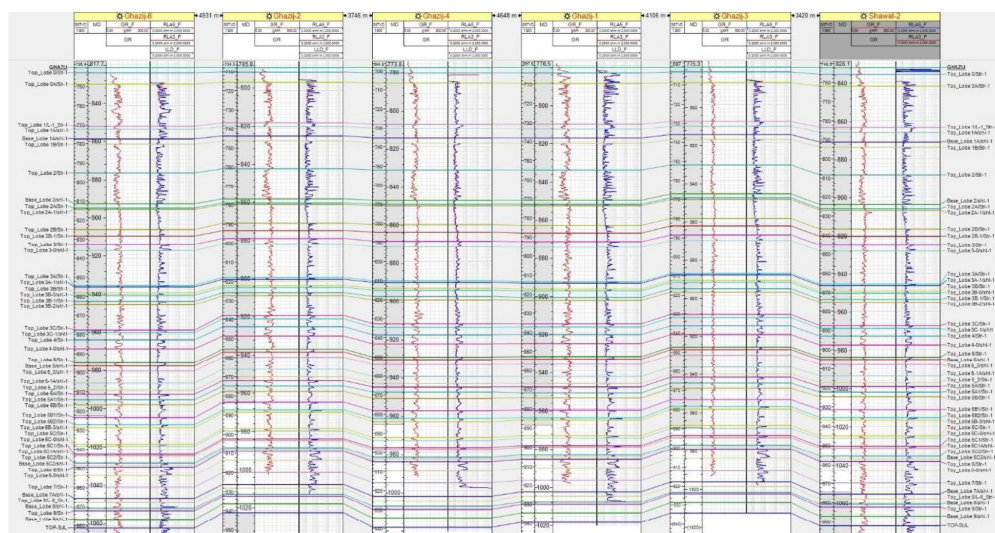


Figure 1—Correlations of limestone Lobes of the Ghazij Formation across the discovery area.

Fracturing Challenge in Thin Laminated Limestone

Interfacial slip is one of the key mechanisms that can influence hydraulic fracture growth, particularly when a propagating fracture intersects bedding planes or natural fractures. In formations with horizontal bedding planes, especially for vertical fractures, fracture height growth is closely tied to the degree of slip along these interfaces. Although in-situ stress contrast is widely recognized as the dominant control on the height containment of a fracture, weak interfaces can also arrest fracture propagation when the effective normal stress across them is low. For horizontal bedding planes, this normal stress corresponds to the vertical overburden stress, which is comparatively lower in shallow formations, increasing the likelihood of slippage and limiting vertical fracture extension.

In an experimental study on fracture containment by layer interfaces (Teufel and Clark, 1984), it was found that the shear strength of an interface was related to the coefficient of friction of the interface and the normal stress applied on the interface. When the interface shear strength is low, the fracture height is contained, and when the shear strength is high, the fracture grows across an interface.

When a hydraulic fracture approaches an interface between contrasting lithologies - such as limestone and shale - its propagation may be altered by mechanisms including delamination, arrest, or penetration across the boundary. Weakly bonded interfaces may facilitate slip or delamination, halting vertical growth as the fracture tip reaches the contact, or redirecting the fracture laterally along the interface. Classic experimental and theoretical studies (He & Hutchinson, 1989; Dundurs, 1969; Daneshy, 1978; Warpinski et al., 1982; Teufel & Clark, 1984; Barree & Winterfeld, 1998; Zhang et al., 2007) have demonstrated that such boundaries, when mechanically weak, act as effective barriers to vertical fracture propagation.

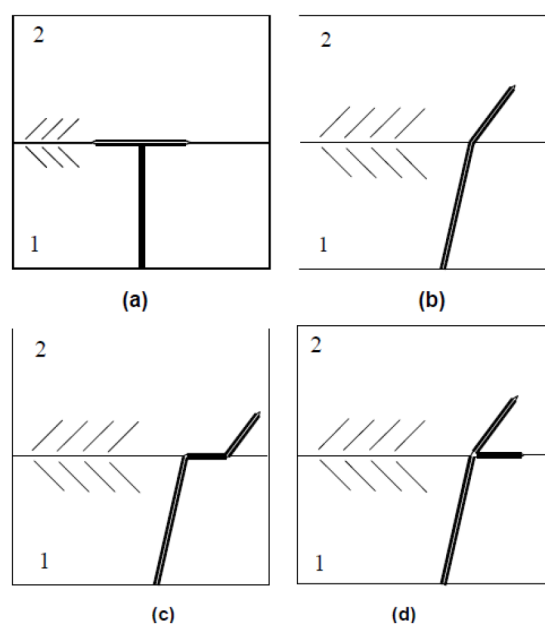


Figure 2—Interaction between fracture and interface: a) delamination, b) kinked penetration, c) jog formation, d) bifurcation.

In thinly laminated sequences, such as the interbedded limestones and shales of the Ghazij reservoir, even moderate contrasts in mechanical properties - tensile strength, Young's modulus, and fracture toughness - can accentuate this effect. Shale layers, acting as mechanical "ceilings" or "floors," may not only impede fracture height growth but also cause deflection, resulting in strata-bound fractures. The presence of these discontinuities increases the likelihood of Mode I tensile fractures transitioning into mixed-mode shear failure, dissipating available fracture energy and confining fractures within individual limestone layers. Consequently, a fracture design in such settings must account for lithological contrasts and interface integrity, adopting perforation and stage-placement strategies that either span the entire laminated interval to

maintain connectivity or are engineered to overcome interface slip, ensuring effective vertical propagation where required.

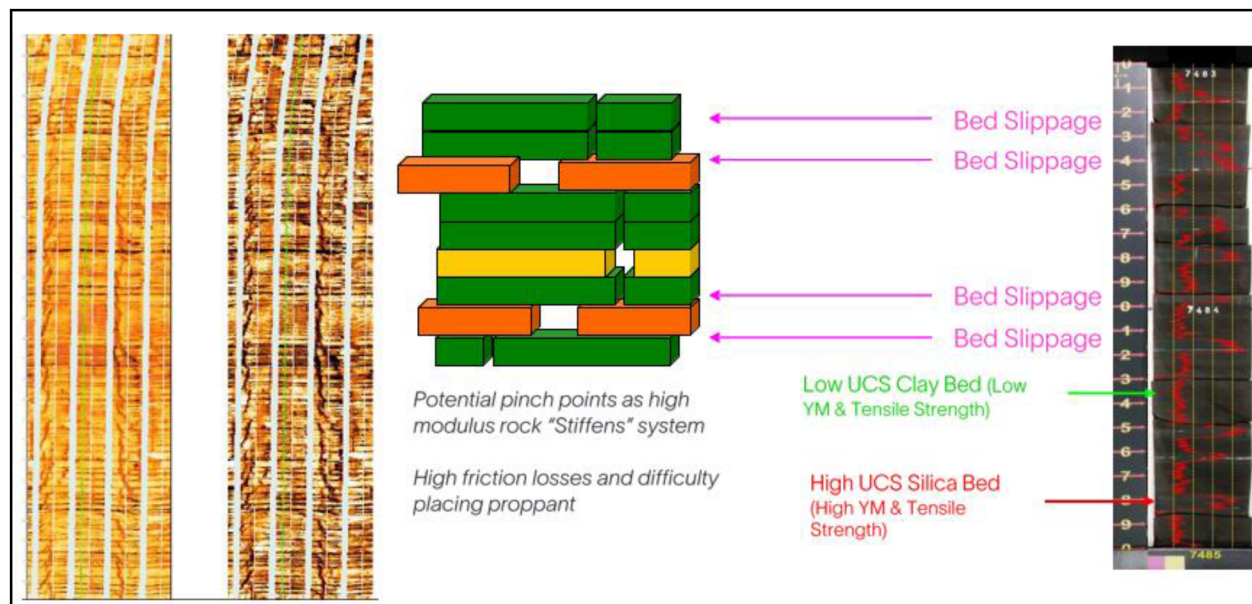


Figure 3—Field geology impact to Fracturing - Lamination.

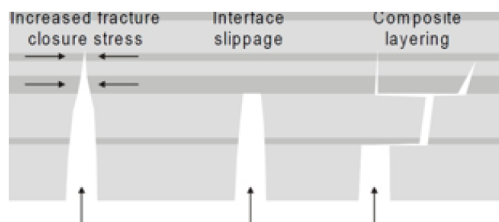


Figure 4—Conceptual picture of three different fracture height confinement mechanisms.

Fracturing Strategy

The fracturing strategy was formulated to evaluate the feasibility of effective fracture initiation and propagation within the laminated limestone-shale intervals of the Ghazij reservoir. The primary objective was to establish hydraulic fracturing as a technically viable stimulation approach for this complex formation. This phase was scoped to demonstrate fracture placement and assess production response. To achieve this, a two-stage hydraulic fracturing was executed using a DST string under rig operation

Stage-1 Perforation Strategy

In Stage-1, the perforation strategy was selected with consideration of the anticipated fracture height containment associated with thin, interbedded limestone-shale sequences in the Ghazij reservoir. Given the possibility of improper fracture propagation and early fracture arrest at shale partings, the decision was made to perforate the entire 20-25 m thick limestone bodies, inclusive of the intervening shale laminae. This approach was intended to ensure complete zone coverage during the fracturing operation, thereby reducing the likelihood that individual sub-layers would remain unstimulated due to lithological heterogeneity. By initiating fractures across the full gross interval, the design aimed to overcome localized mechanical barriers within the sequence and promote more uniform fracture growth across the vertical section.

In addition, perforating the entire composite interval was anticipated to minimize the barrier effect of thin shale beds situated mid-section within the limestone packages, which could otherwise segment the fracture network. This comprehensive coverage was expected to maintain hydraulic connectivity between all sub-

units and the wellbore, facilitating fluid entry into both primary and secondary fracture planes. The strategy was also designed to mitigate the risk of production loss from stimulated zones due to shale pinching, where isolated fracture compartments might otherwise be cut off from the wellbore. By adopting this perforation scheme, Stage-1 operations aimed to maximize stimulated rock volume and ensure efficient fluid injection throughout the mechanically complex, laminated carbonate-shale reservoir.

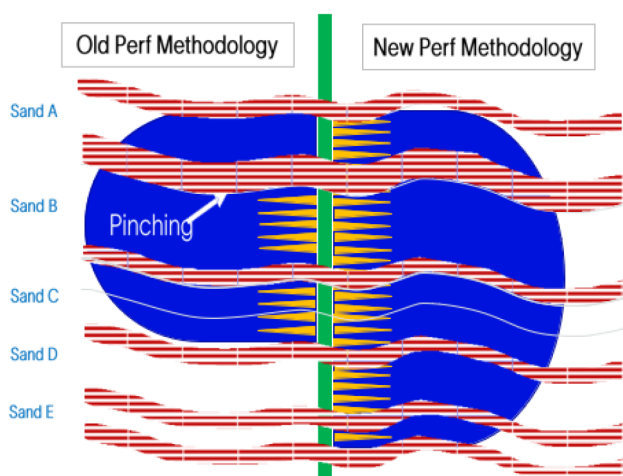


Figure 5—Field geology impact to Fracturing - Perforation strategy

Stage-1 Treatment:

Diagnostic Testing & Analysis

A series of diagnostic tests, including the Breakdown Injection Test (BDI), Step-Rate Test (SRT), Step-Down Test (SDT), and mini-frac calibration, were executed to establish key fracture parameters, assess near-wellbore conditions, and calibrate the 1D mechanical earth model (MEM) for the Ghazij formation. The BDI indicated a BH ISIP of 2,818 psi (0.87 psi/ft) and a closure pressure of 2,571 psi (0.79 psi/ft), both below the overburden gradient, confirming a normal stress regime and a minimal risk of T-shaped fractures. High leak-off was evident by a fluid efficiency of 7.6%. The SRT/SDT sequence identified a fracture extension pressure of 2,927 psi and low tortuosity friction (~260 psi), validating stable fracture propagation while revealing significant frictional pressure losses in the 2.25" tubing. The mini-frac, using the crosslinked gel, demonstrated a sharp improvement in fluid efficiency (73%), net pressure gain (517 psi), and fracture containment, though temperature log anomalies between 980-997 m suggested interfacial slippage at lithological boundaries. This slippage, likely due to stress contrast between limestone and shale layers, implied potential fracture height arrest or lateral diversion along bedding planes, making perforation placement critical to avoid stimulating isolated intervals. Temperature log shows most of the fluid is injected in the top 5m interval of the perforation whereas small deflection in temperature is evident in bottom intervals as well. This shows that the fracture initiation is from top 5m interval where temperature deflection is maximum and most of the fluid is injected there.

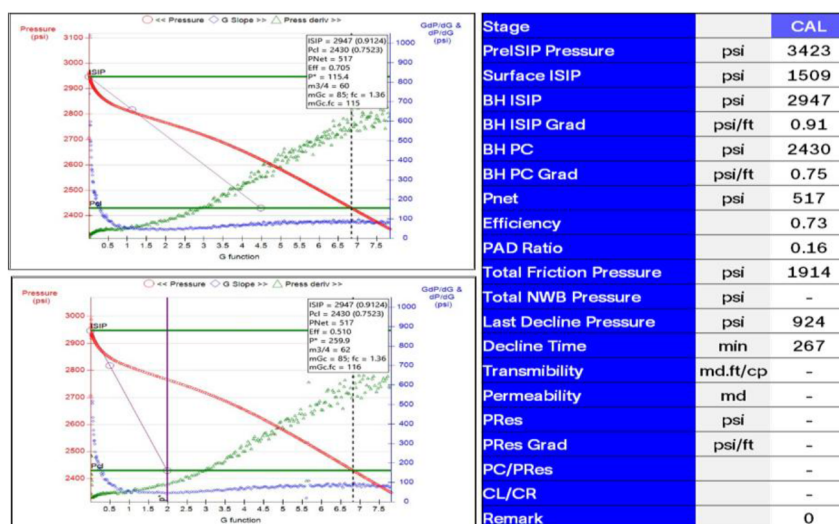


Figure 6—Stage-1 DFIT Analysis

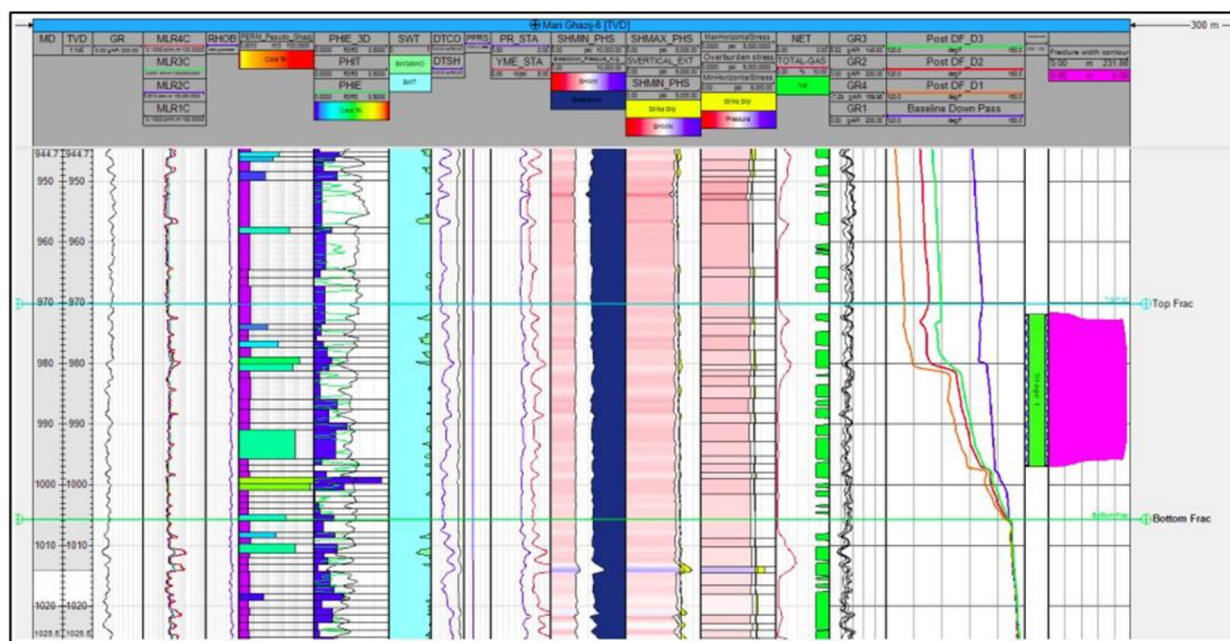


Figure 7—Stage-1 Temperature Survey Plot

Stage-1 Main Frac Treatment

The Stage-1 main frac was designed with a conservative proppant ramp-up (0.5 PPA to 6.0 PPA) to mitigate screen-out risks while managing leak-off and interfacial slip tendencies. A total of 2,302.5 bbls of slurry and 154,986 lbs of ceramic proppant (40/70, 30/50, 20/40 blends) were placed efficiently. Treating pressures remained stable (avg. 3,262 psi), and pad volume was maintained at a 0.32 ratio. Post-frac modeling indicated a frac height of ~90 m - exceeding the 30 m target - confirming height growth dominance. However, fracture width averaged only 0.10 in (vs. 0.18 in design), reducing conductivity by 33%.

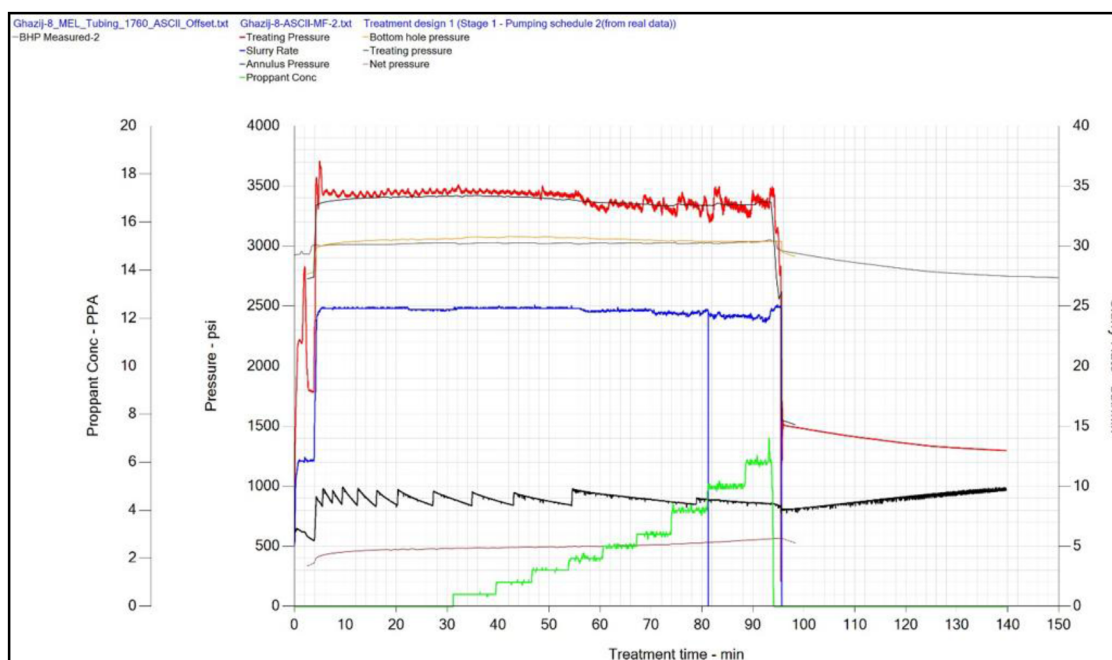


Figure 8—Stage-1 Net pressure matching plot

Post-Frac Performance:

Flowback operations revealed slow cleanup and no production gains, with gas rate averaging 0.9 MMSCFD and no proppant flowback. The delayed fluid recovery was attributed to a possible capillary imbibition and the reduced fracture width, which limited the effective drainage area despite increased stimulated rock volume. The absence of a significant productivity boost reinforced the diagnostic findings - fracture geometry was height-biased with restricted width, possibly due to large number of perforations.

Key Lessons Learned in the Context of Interfacial Slippage & Perforation Strategy

The net pressure matching, integrated testing, temperature logging, and production logging data collectively point toward fracture height growth as most likely contributor to Stage-1 underperformance. The temperature log during the main frac showed that the uppermost 5 m of the 25 m perforated interval was the primary fluid intake and fracture initiation point, with only minor thermal deflections observed in the lower intervals. This indicates that while the fracture initiated in the top section, fracture energy was dissipated into the other intervals, suggesting dissipation of available fracture energy beyond the main intake zone. The absence of any increase in net pressure compared to the mini-frac, combined with very stable pressure trends, supports this understanding and implies limited fracture width development despite the larger treatment volume. Post-job modeling further confirmed a narrower fracture width than design expectations, consistent with the energy diversion away from the width creation and possibly into the height propagation. These observations suggest that while interbedded lithologies and potential interface weaknesses may not have been as dominant containment barrier as anticipated, the resulting geometry was non-optimal - favoring height growth at the expense of width and conductivity - ultimately reducing overall stimulation effectiveness.

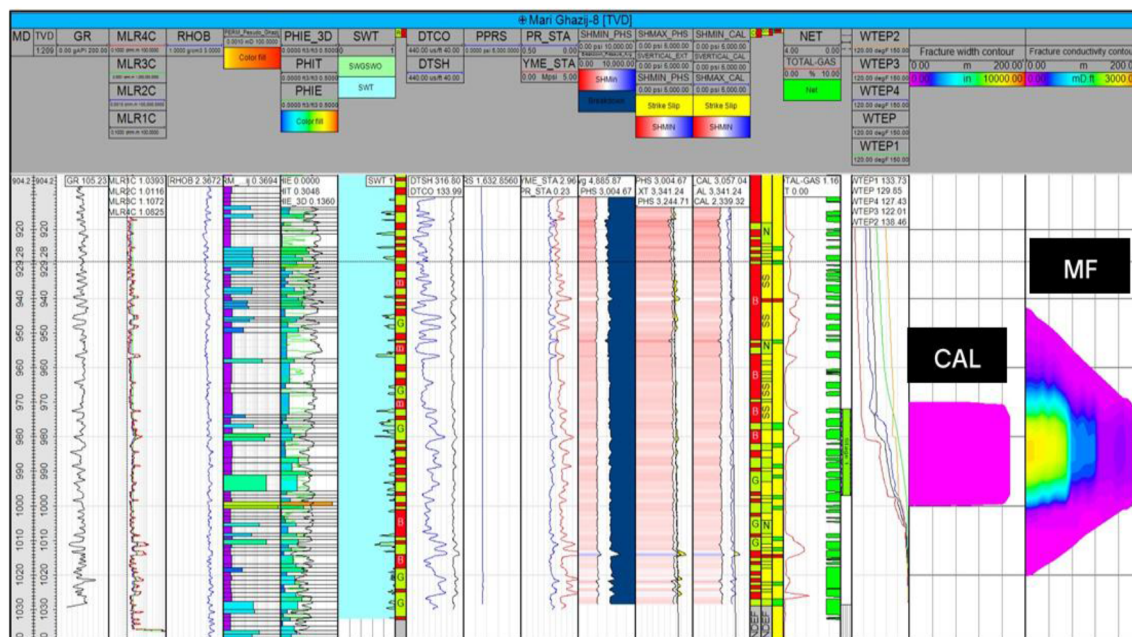


Figure 9—Stage-1 Frac Geometry Plot

Stage-2 Treatment:

Perforation Interval Selection - Structured Approach

For Stage-2, the operator adopted an engineered perforation selection strategy based on the lessons learned from Stage-1, applying a systematic workflow that integrates both Reservoir Quality (RQ) and Completion Quality (CQ) to improve stimulation efficiency.

RQ was defined as the interval's inherent capacity to store and deliver hydrocarbons, primarily governed by parameters such as gas-in-place, total and effective porosity, hydrocarbon saturation, permeability, and pore pressure. While RQ is a fixed reservoir property that cannot be altered, its accurate assessment in thinly-aminated formations such as the Ghazij is challenging due to the vertical resolution limits of conventional open-hole logging. To mitigate this, the operator relied heavily on the image log interpretation to identify clean, continuous limestone intervals and incorporated production logging results from offset wells to confirm the contributing zones. This ensured that perforation selection targeted high-quality pay while minimizing exposure to non-productive shale layers. The overarching goal was to define the smallest practical perforation interval to focus fracture initiation, concentrate fracture energy, and maximize fracture width within the most productive zones, thereby avoiding unnecessary dissipation of energy into mechanically weak or non-reservoir intervals.

CQ was evaluated as the ability to create an effective fracture network with sustained conductivity, primarily influenced by the in-situ stress conditions and the key rock mechanical properties such as Young's modulus, unconfined compressive strength (UCS), and Poisson's ratio.

The engineered workflow began with constructing RQ and CQ indices, quantified through weighted integration of multiple datasets - RQ derived from petrophysical data including open hole logs, and cased hole logs whereas CQ determined using outputs from a one-dimensional mechanical earth model (MEM) that incorporated elastic property profiles, stress orientation, and anisotropy. Screening started with the identification of intervals exhibiting high RQ, after which CQ analysis was applied to those zones to refine the selection.

	Criteria	Value
RQ	V Shale	<0.25 v/v
	Resistivity	>0.1 ohm.m
	Water Saturation	<0.75
	Porosity	>10%
CQ	SHMin	<3000 psi
CCQ	Stress Regime	Normal, Strike-Slip, Reverse

Figure 10—Reservoir Quality, Completion Quality, Completion Class Quality Parameters Table

In Stage-2 planning, particular attention was given to ensure that the selected perforation interval exhibited relatively uniform effective stresses across a 3-8 m interval, combined with the mechanical properties favorable for the fracture initiation - specifically lower UCS values to reduce breakdown pressure requirements and higher Young's modulus to maintain fracture aperture and conductivity under closure stress.



Figure 11—Stage-2 Perforation Intervals

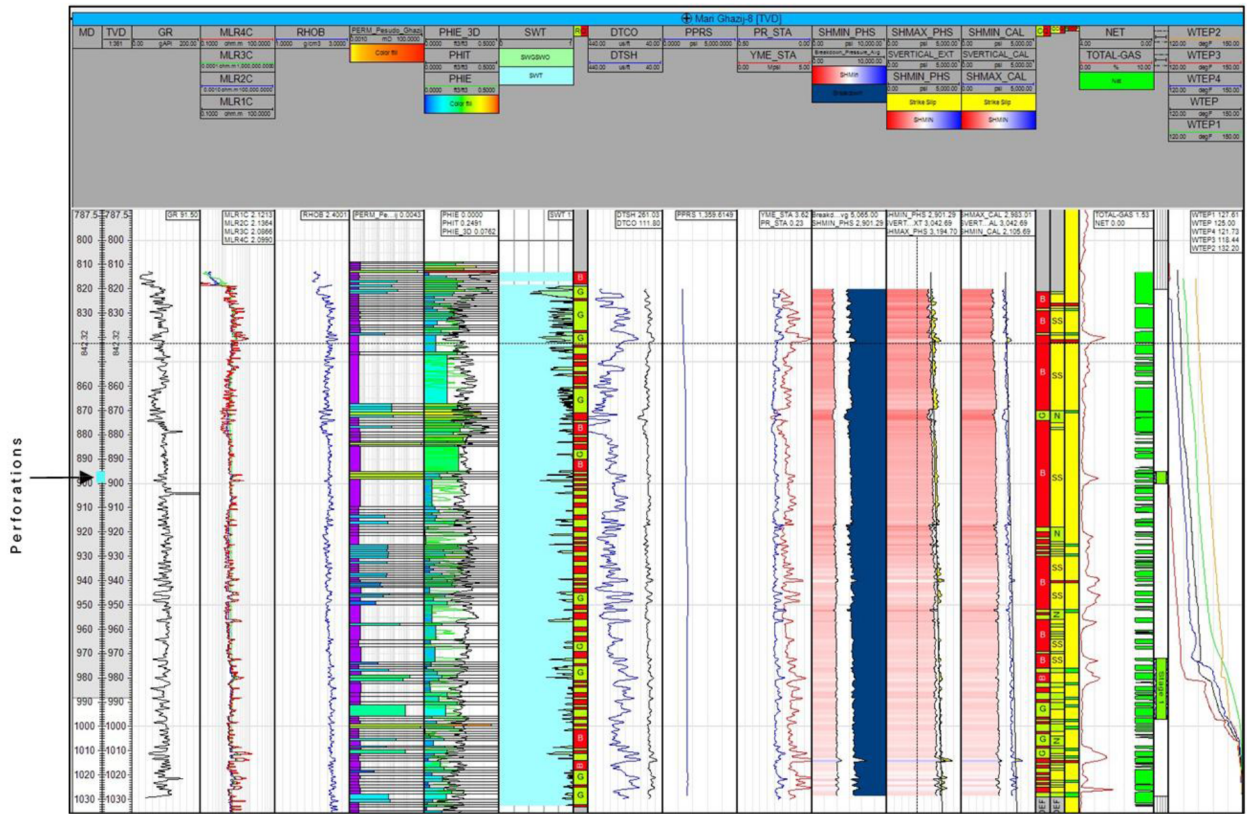


Figure 12—Stage-1 Temperature Survey Plot

Stage-2 Treatment:

Stage-2 began with a Breakdown Injection Test (BDI) designed to calibrate the main treatment design. The breakdown was achieved after 20 bbls at a maximum surface pressure of 2,379 psi. The injection proceeded at maximum rate of 5.2 bbl/min, confirming stable injection and low near-wellbore friction (80 psi). Pressure analysis indicated an ISIP gradient of 0.83 psi/ft - slightly lower than that in Stage-1 - suggesting minor stress shadowing. Closure was observed at 1,860 psi (0.63 psi/ft) lower than that in Stage-1, implying reduced fracture complexity. Fall-off analysis showed no height growth signatures, contrasting with that in Stage-1, and resulted in guided-pad volume adjustment along with proppant schedule refinement.

Well	Ghazij-8- Ghazij-8-	
	1	2
DFIT		
PrelSIP Pressure	psi	1952 1582
Surface ISIP	psi	1380 1214
BH ISIP	psi	2818 2449
BH ISIP Grad	psi/ft	0.87 0.87
BH PC	psi	2571 1860
BH PC Grad	psi/ft	0.87 0.63
Pnet	psi	247 589
Efficiency		0.18 0.18
PAD Ratio		- 0
Total Friction Pressure	psi	572 368
Total NWB Pressure	psi	- 80
Last Decline Pressure	psi	169 284
Decline Time	min	123 108
Transmissibility	md.ft/cp	- 133
Permeability	md	- 1.862
PRes	psi	1582 1380
PRes Grad	psi/ft	0.48 0.48
PC/PRes		1.66 1.38
CL/CR		1.03 0.63
Remark		0 0

Figure 13—Stage-1&2 DFIT Analysis Comparison

The redesigned main treatment targeted improved fracture width, conductivity, and containment. Pad volume was set at 20% of total slurry, split equally between HVFR and the crosslinked fluid for the fracture geometry control before the proppant arrival. A single proppant type (20/40) was selected to maximize the conductivity, with its aggressive concentration ramp-up from 1.0 to 8.0 PPA after pumping 2-2.5 wellbore volumes, along with the contingency of 910 PPA if pressure trends allowed. The total proppant target was 200,000 lbs, with 94% expected placement in the fracture network. Breaker was introduced early in the flush to accelerate gel break, with increased loading and extended shut-in time (15+ hours) to address pressure-dependent leak-off.

Execution of the main treatment involved 1,485 bbl slurry pumped at an average 24.2 bbl/min (peak 25.3) with maximum surface pressure of 5,459 psi prior to screen out. Initial HVFR and crosslinked pads totaled 281 bbl, followed by staged proppant ramp-up to 8.0 PPA. Placement totaled 193,015 lbs of 20/40 proppant, with 37% pumped at 5.0 PPA and final stages sustaining 7.0-8.0 PPA. Proppant left in the wellbore was only 10,944 lbs (5.7%), demonstrating excellent placement efficiency and validating the high concentration pumping strategy.

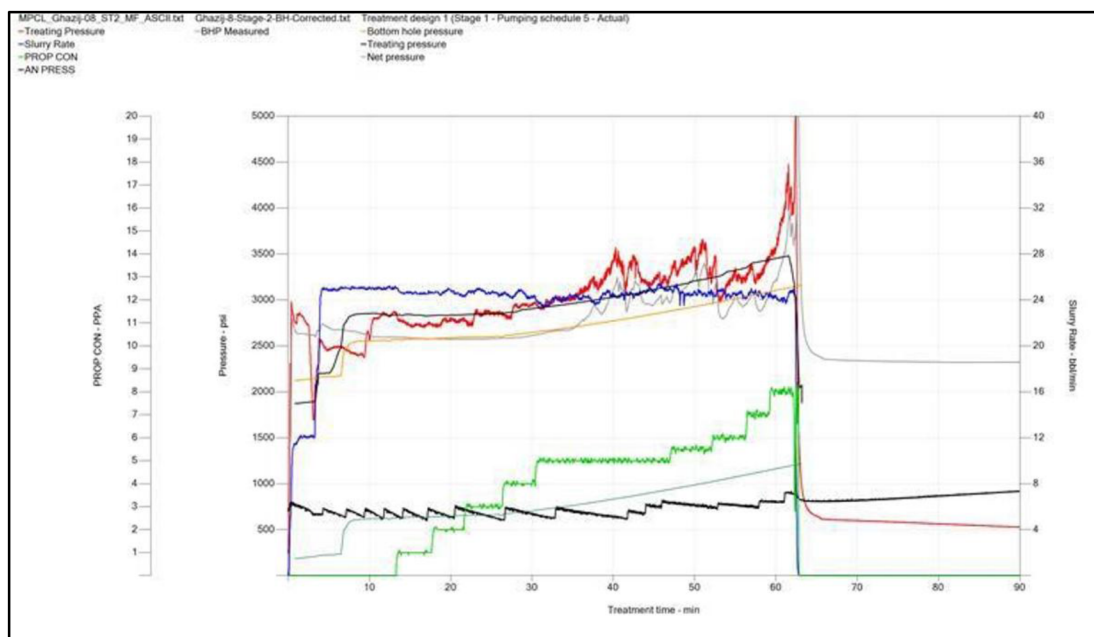


Figure 14—Stage-2 Net Pressure Matching Plot

Post-treatment modeling showed a propped half-length of 165.6 m - slightly shorter than Stage-1 - but with improved height containment (29.0 m vs 90.0 m), reflecting better fracture confinement. Average propped width increased to 0.18 inches (vs 0.10 in Stage-1), attributable to the limited perforation intervals, higher proppant concentrations and the optimized fluid design. Net pressure development improved by 112% (1,200 psi vs 567), indicating more effective energy transfer into fracture propagation. Conductivity increased to 2,058 mdft, supported by a higher average proppant loading (>4.0 PPA vs 2.6 in Stage-1) and a better proppant retention in the target interval. Pressure matching confirmed a superior management of the pressure response and the fracture geometry.

Clean-up operations demonstrated significant production uplift. Pre-frac rates of 0.8 MMSCFD increased to 3.18

MMSCFD on a 128/64" choke with stable flowing pressures (62-73 psig) and temperatures (82-90°F). Reducing the choke to 32/64" resulted in WHFP of 435-460 psig with sustained 2.37-2.43 MMSCFD, confirming robust fracture conductivity and reservoir deliverability.

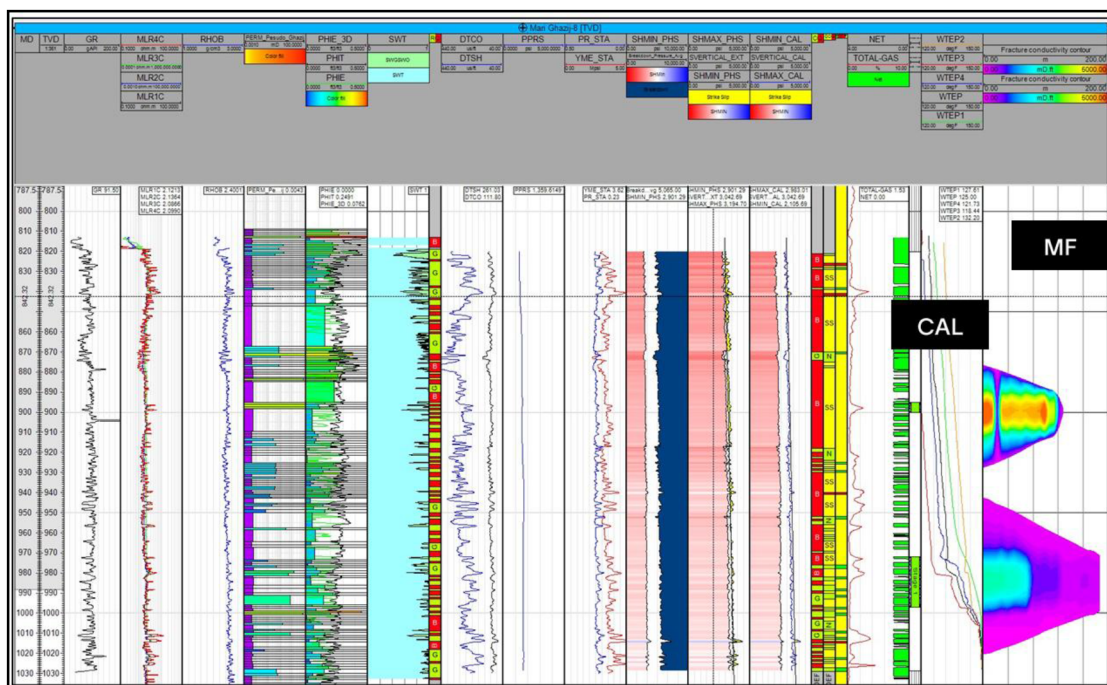


Figure 15—Stage-2 Frac Geometry Plot

Comparative analysis against pre-frac nodal modeling showed that the treatment overcame significant formation damage (skin 3.2) that acidizing alone would not have removed—acid stimulation was estimated to yield only ~1.5 MMSCFD under similar conditions. The Stage-2 fracture achieved over twice that rate, outperforming offset treatments and demonstrating the strategic value of aggressive but controlled proppant placement in the Ghazij formation.

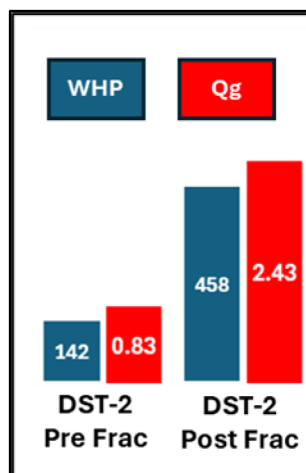


Figure 16—Stage-2 Testing Results (Pre & Post Frac)

Post-frac performance confirmed nearly 300% production increase, setting a new benchmark for Ghazij stimulation and providing a replicable template for future stages.

Conclusion

By balancing RQ-driven productivity potential with CQ-driven fracture efficiency, the operator overcomes the height- growth and fracture-width limitations observed in Stage-1, where excessive fluid intake into

a single upper interval and probable energy dissipation across bedding planes had hindered geometry optimization.

Acknowledgments

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List of Abbreviations:

TSO	Tip Screen Out
DFIT	Diagnostic Fracture Injection Test
DC	Drill Collars
1D MEM	One Dimensional Mechanical Earth Model
ISIP	Instantaneous Shut-in Pressure
SRT	Step Rate Test
SDT	Step Down Test
SIWHP	Shut-in Well Head Pressure
BPM	Barrels Per Minute
ppt	pounds per thousand gallons of fluid
HvFR	High Viscosity Friction Reducer
Pnet	Net Pressure
MMSCFD	Millions Standard Cubic Feet per Day

References:

- Miskimins, J. L., & Barree, R. D. (2003, March 22-25). Modeling of hydraulic fracture height containment in laminated sand and shale sequences (SPE-80935-MS). Paper presented at the SPE Production and Operations Symposium, Oklahoma City, OK. Society of Petroleum Engineers. <https://doi.org/10.2118/80935-MS>
- Daneshy, A. A. (1977, October 9-12). Hydraulic fracture propagation in layered formations (SPE-6088-MS). Paper presented at the SPE-AIME 51st Annual Fall Technical Conference and Exhibition, New Orleans, LA. Society of Petroleum Engineers. <https://doi.org/10.2118/6088-MS>
- Teufel, L. W., & Clark, J. A. (1984). Hydraulic fracture propagation in layered rock: Experimental studies of fracture containment. *SPE Journal*, **24**(1), 19–32. <https://doi.org/10.2118/9870-PA>
- He, M. Y., & Hutchinson, J. W. (1989). Crack deflection at an interface between dissimilar elastic materials. *International Journal of Solids and Structures*, **25**(9), 1053–1067. [https://doi.org/10.1016/0020-7683\(89\)90021-8](https://doi.org/10.1016/0020-7683(89)90021-8)
- Dundurs, J. (1969). Discussion of paper: Edge-bonded dissimilar orthogonal elastic wedges under normal and shear loading. *Journal of Applied Mechanics*, **36**(3), 650. <https://doi.org/10.1115/L3564739>
- Daneshy, A. A. (1978). Hydraulic fracture propagation in layered formations. *SPE Journal*, **18**(1), 33–41. <https://doi.org/10.2118/6752-PA>
- Warpinski, N. R., Schmidt, R. A., & Northrop, D. A. (1982). In situ stresses: The predominant influence on hydraulic fracture containment. *Journal of Petroleum Technology*, **34**(3), 653–664. <https://doi.org/10.2118/8932-PA>
- Barree, R. D., & Winterfeld, P. H. (1998, September 27-30). Effects of shear planes and interfacial slip on fracture growth and treating pressures (SPE-48926-MS). Paper presented at the SPE Annual Technical Conference and Exhibition, New Orleans, LA. Society of Petroleum Engineers. <https://doi.org/10.2118/48926-MS>
- Zhang, X., Jeffrey, R. G., & Thiercelin, M. (2007, January 29-31). Effects of frictional geological discontinuities on hydraulic fracture propagation (SPE-106111-MS). Paper presented at the SPE Hydraulic Fracturing Technology Conference, College Station, TX. Society of Petroleum Engineers. <https://doi.org/10.2118/106111-MS>